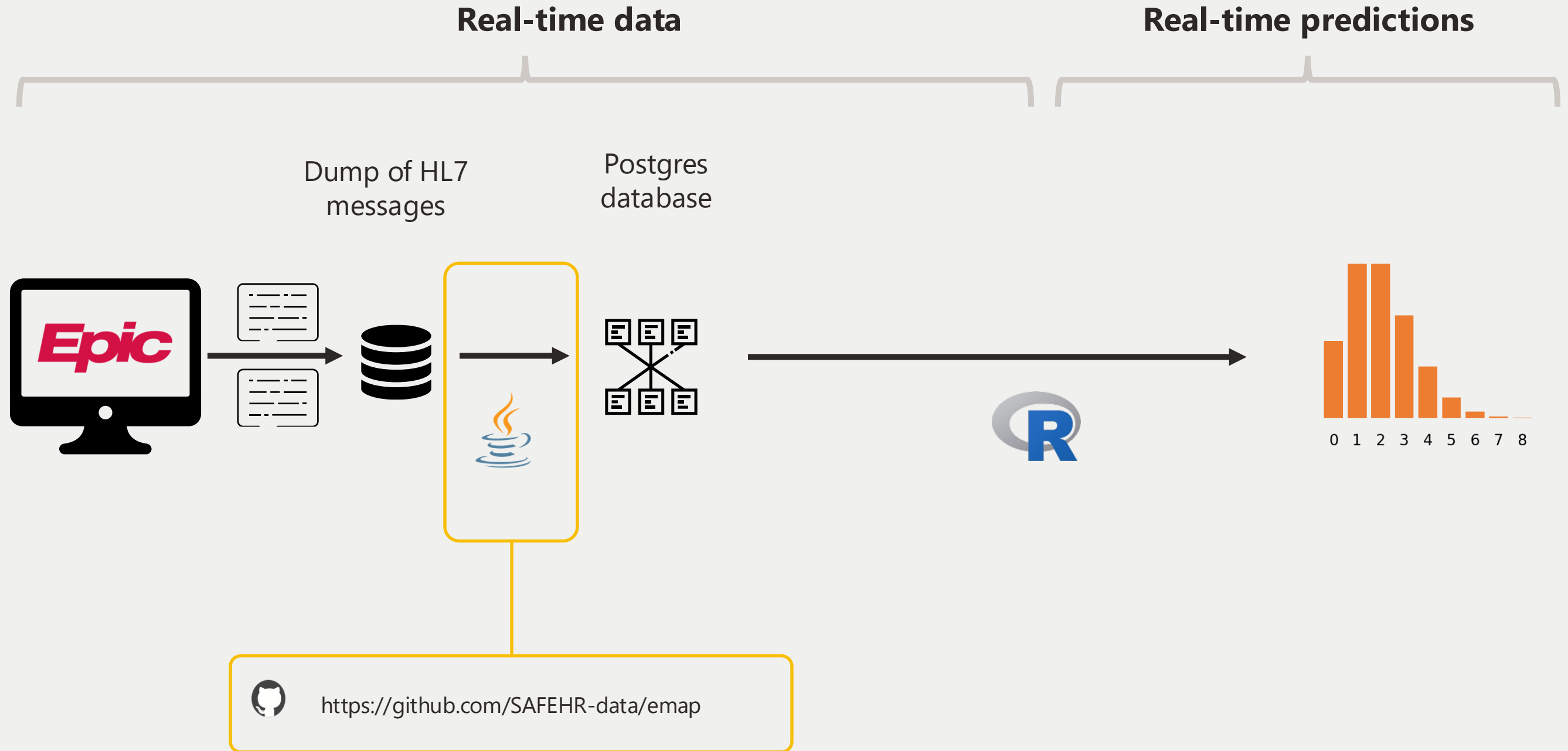


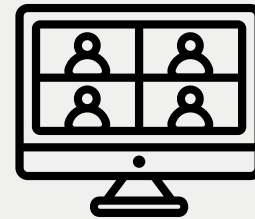
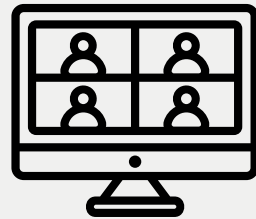
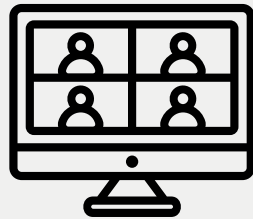
The story of patientflow: From hospital prototype to Python package

Dr Zella King, Senior Research Associate

Clinical Operational Research Unit, UCL
and University College London Hospital

From hospital prototype ...





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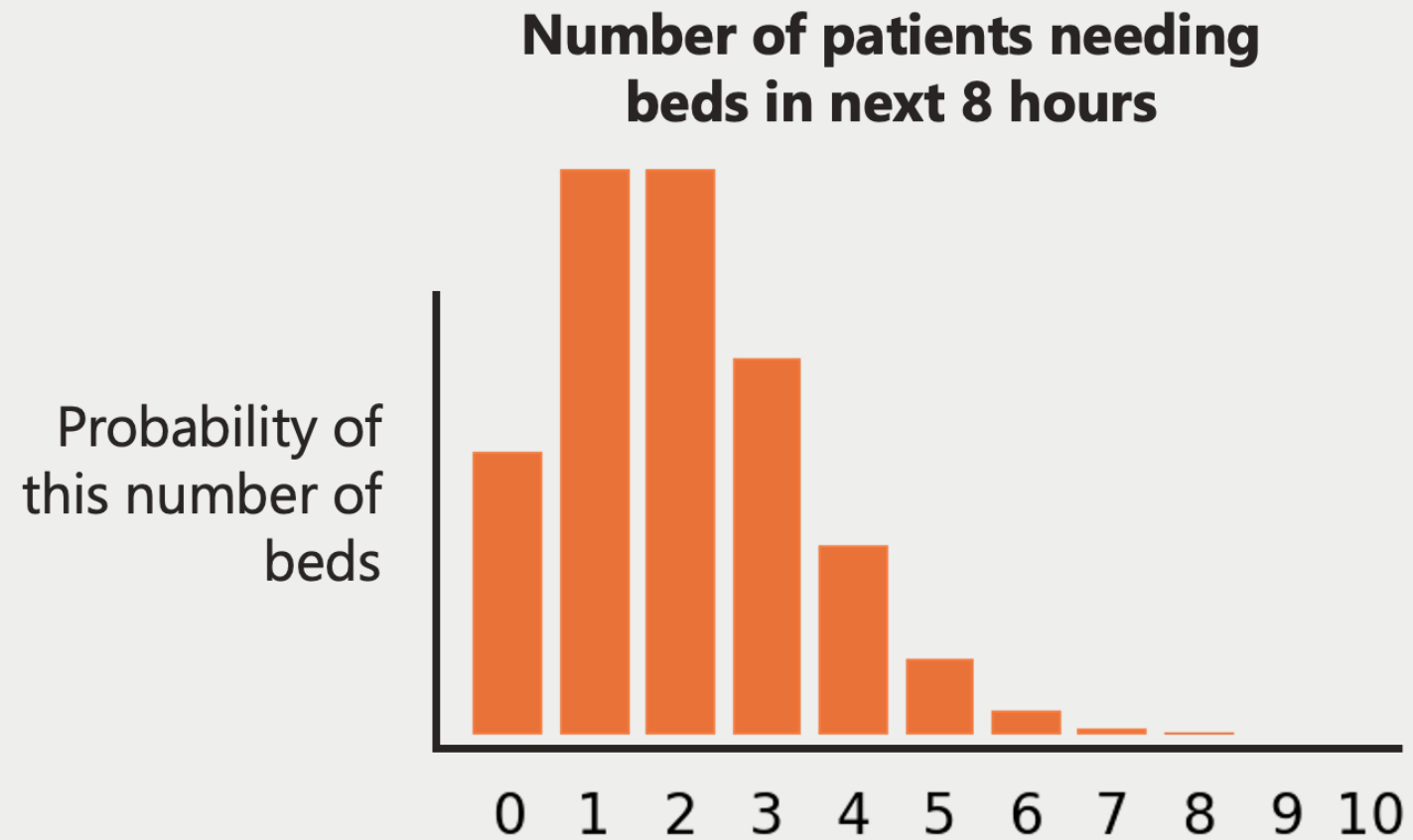
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+10

+1

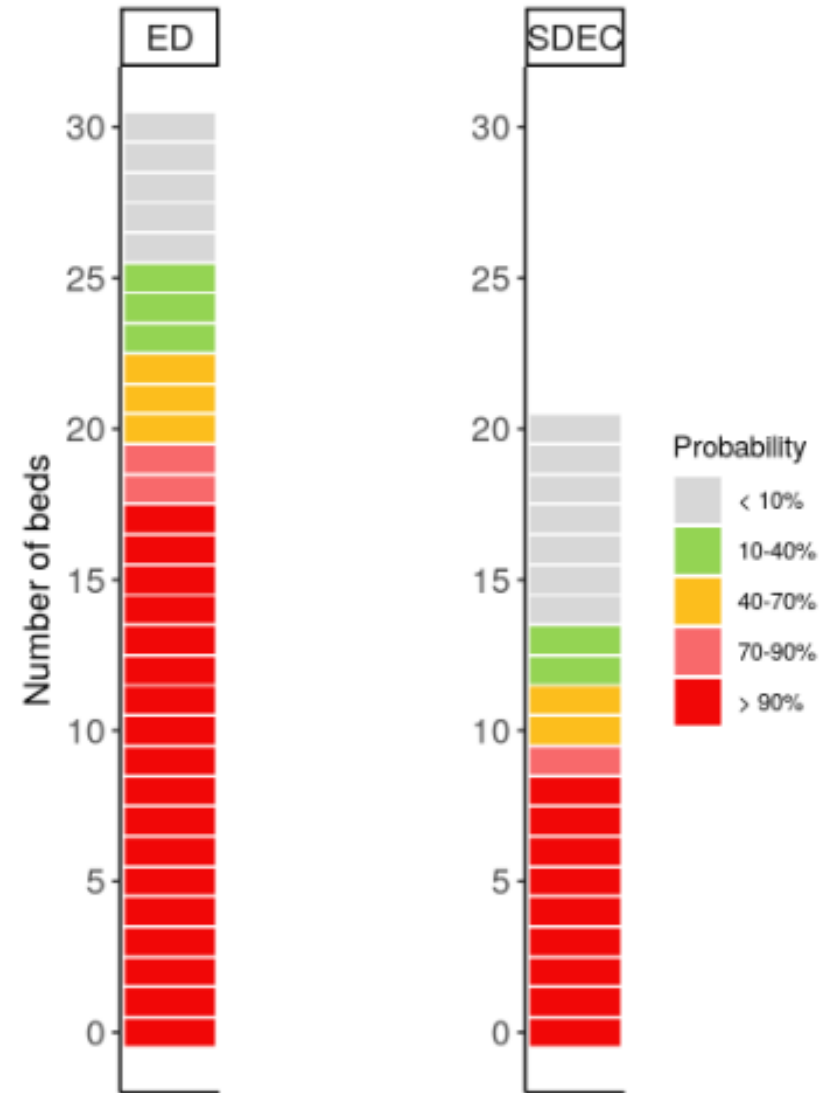
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Beds needed at some point for patients currently in ED & SDEC

Excludes beds for patients under 18, and patients marked as OTF



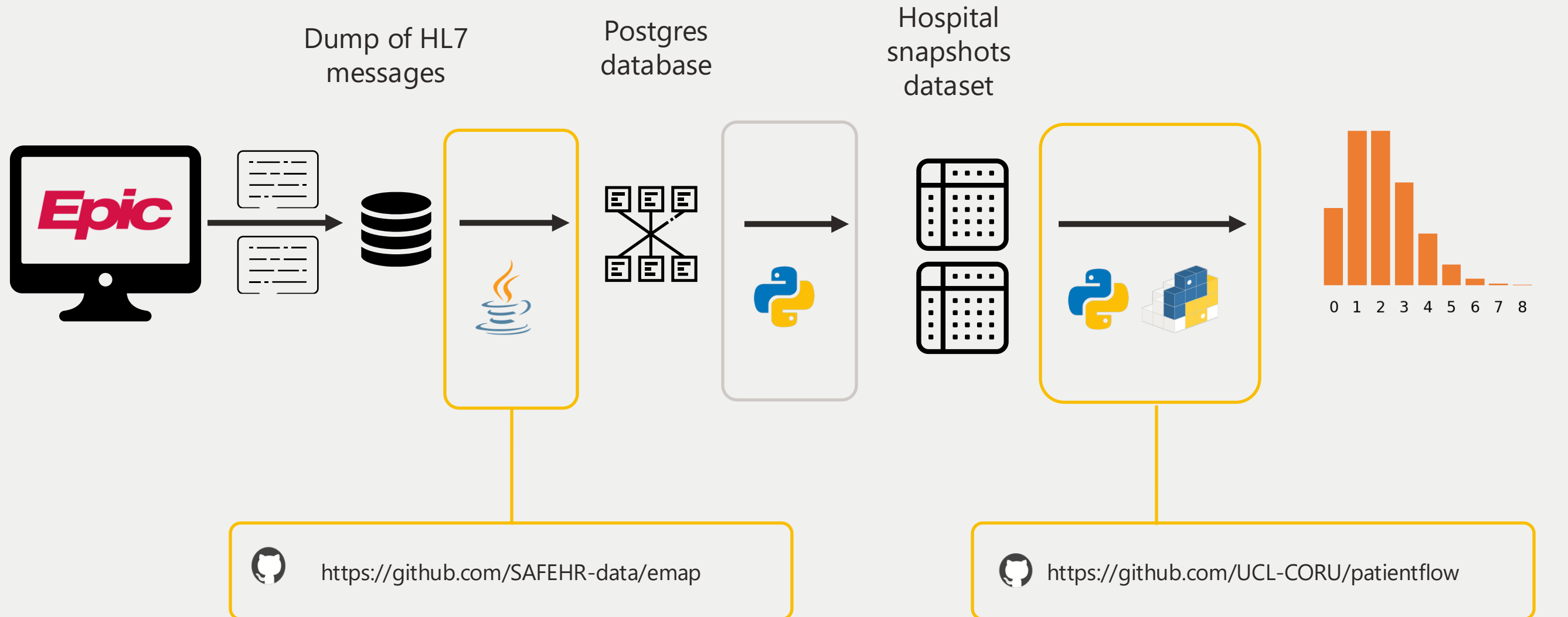
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Nature
Digital
Medicine

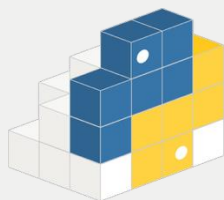


... to Python package

Real-time data

Real-time predictions





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patientflow 1.1.0



[Latest version](#)

```
pip install patientflow==1.1.0
```



Released: Oct 6, 2025

Code and explanatory notebooks for predicting short-term hospital bed capacity using real-time data

Navigation

 [Project description](#)

 [Release history](#)

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Maintainers



zmek

Unverified details

Project description

patientflow: a Python package for real-time predictions of hospital bed demand from current and incoming patients

 pre-commit  enabled  Tests  passing  Linting  passing  Documentation  passing docs [ucl-coru.github.io](#) License  MIT

 pypi  v1.1.0  python  3.10 |  3.11 |  3.12 |  3.13 DOI [10.5281/zenodo.15722296](#)

 ORCID [0000-0001-7389-1527](#)  ORCID [0009-0007-4110-7284](#)  ORCID [0000-0001-9928-1516](#)  ORCID [0000-0001-9104-7960](#)
 ORCID [0000-0003-1882-5476](#)

Summary

`patientflow`, a Python package for real-time prediction of hospital bed demand from current and incoming patients, creates output that is useful for bed managers in hospitals, allowing researchers to easily develop predictive models and demonstrate their utility to practitioners.





README MIT license



patientflow: a Python package for real-time predictions of hospital bed demand from current and incoming patients

pre-commit enabled Tests passing Linting passing Documentation passing docs ucl-coru.github.io

License MIT pypi v1.1.0 python 3.10 | 3.11 | 3.12 | 3.13 DOI 10.5281/zenodo.15722296

ORCID 0000-0001-7389-1527 ORCID 0009-0007-4110-7284 ORCID 0000-0001-9928-1516

ORCID 0000-0001-9104-7960 ORCID 0000-0003-1882-5476

Summary

`patientflow`, a Python package for real-time prediction of hospital bed demand from current and incoming patients, creates output that is useful for bed managers in hospitals, allowing researchers to easily develop predictive models and demonstrate their utility to practitioners.

We originally developed this code for University College London Hospitals (UCLH) to predict the number of emergency admissions they should expect within the next eight hours. Our method used real-time data from their Electronic Health Record (EHR) system. We wrote code to convert patient-level data, extracted from the EHR at a point in time, into predicted numbers of admissions in the following hours. We also wrote code to help us evaluate the predictions.






PatientFlow




zmek/patientflow
v1.1.0 ☆ 30 🍴 2

PatientFlow
patientflow: a Python package for real-time predictions of hospital bed demand from current and incoming patients
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[API reference](#)
Notebooks >
Src >

API reference

PatientFlow: A package for predicting short-term hospital bed demand.

This package provides tools and models for analysing patient flow data and making predictions about emergency demand, elective demand, and hospital discharges.

aggregate

Aggregate Prediction From Patient-Level Probabilities

This submodule provides functions to aggregate patient-level predicted probabilities into a probability distribution using a generating function approach.

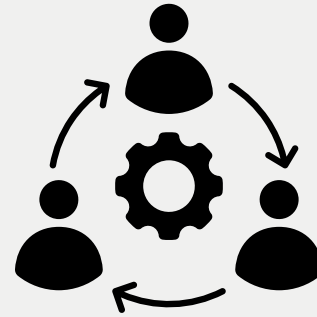
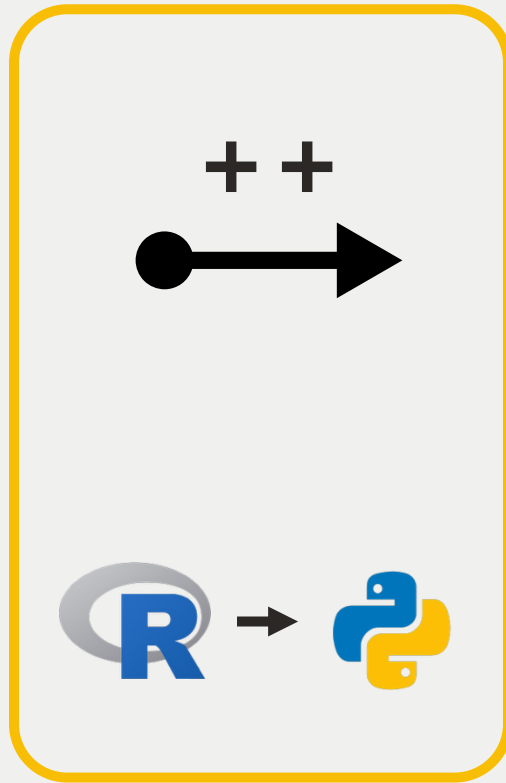
The module uses dynamic programming to compute the exact probability distribution of sums of independent Bernoulli random variables, which is mathematically equivalent to multiplying their generating functions but computationally much more efficient.

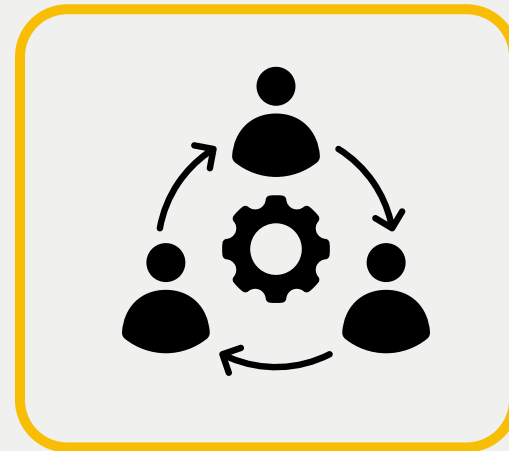
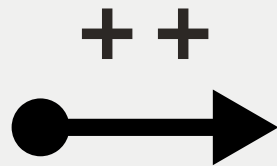
Functions:

Name	Description
<code>BernoulliGeneratingFunction</code> : class	Efficient generating function implementation for sums of Bernoulli random variables.

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patientflow
aggregate
 BernoulliGeneratingFunction
 exact_distribution
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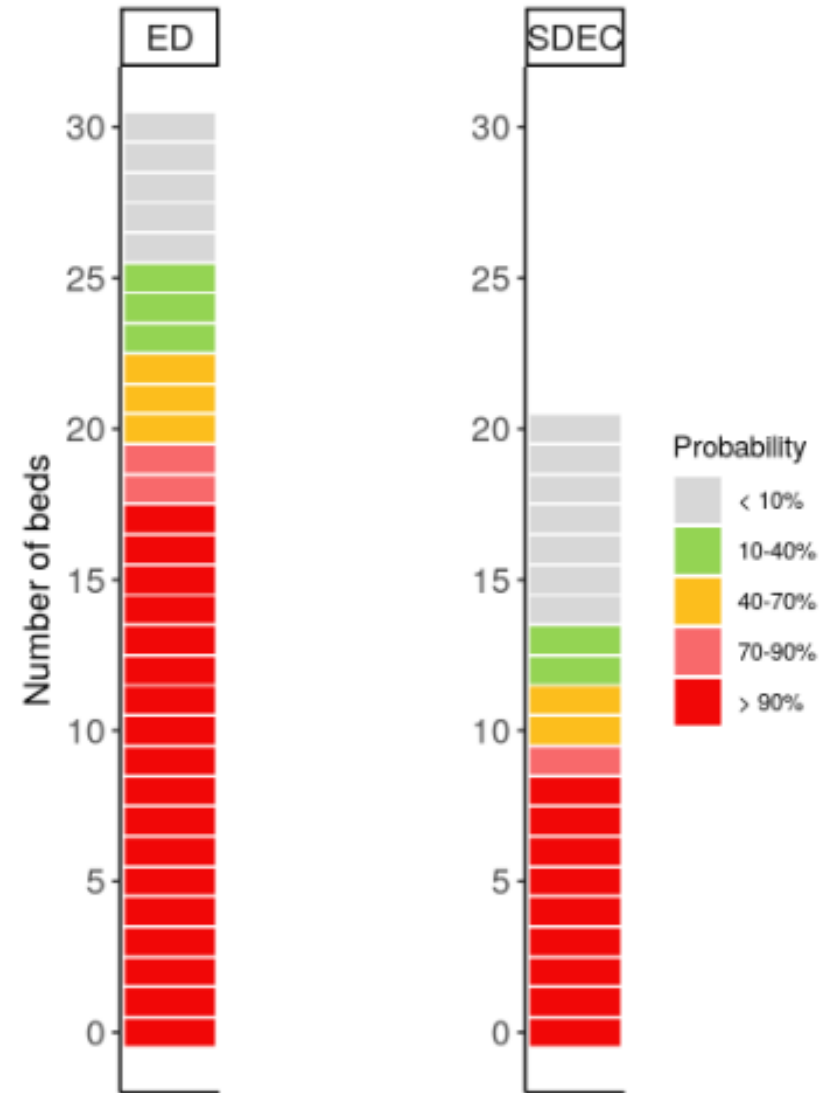
The story





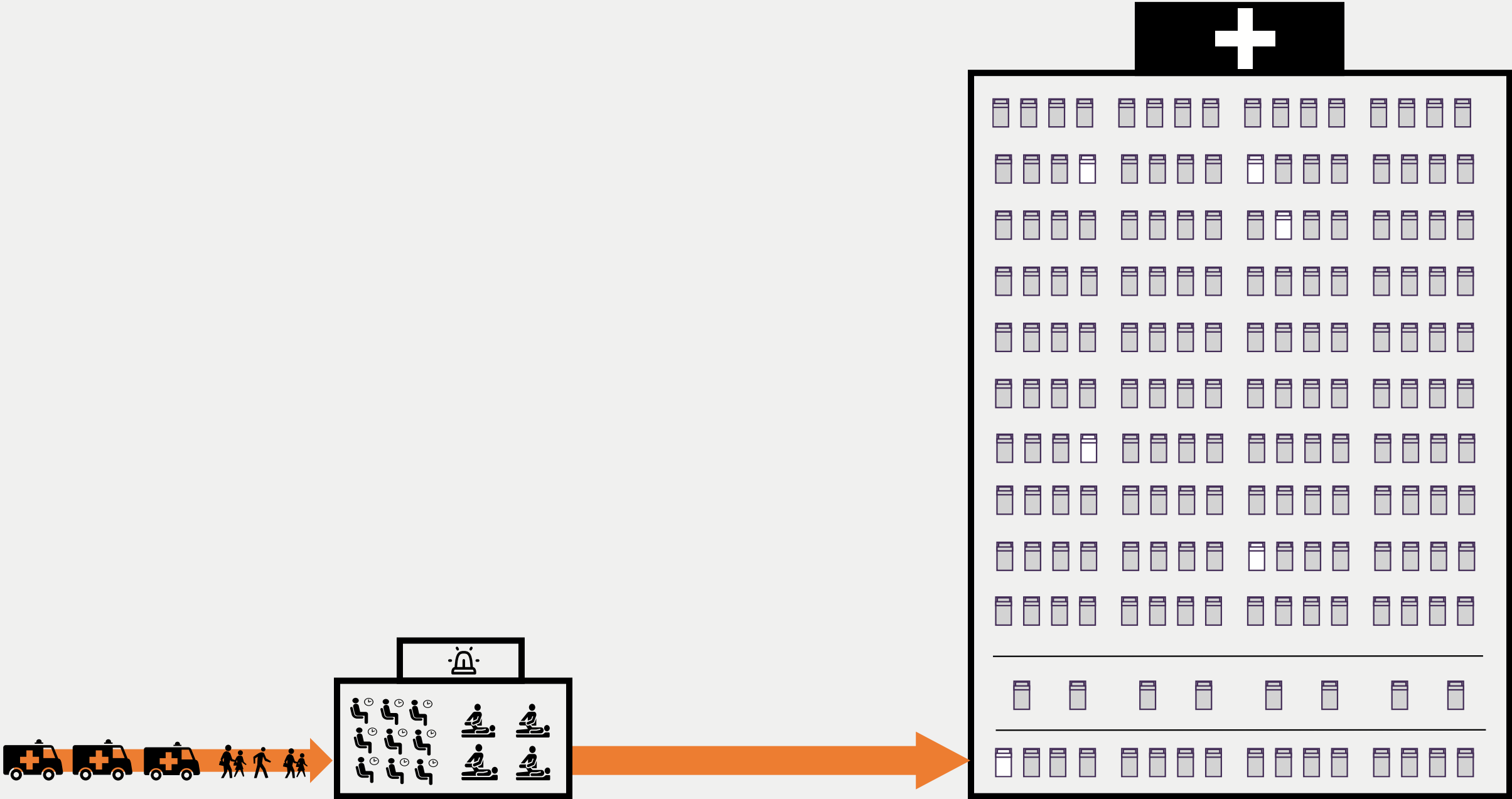
Beds needed at some point for patients currently in ED & SDEC

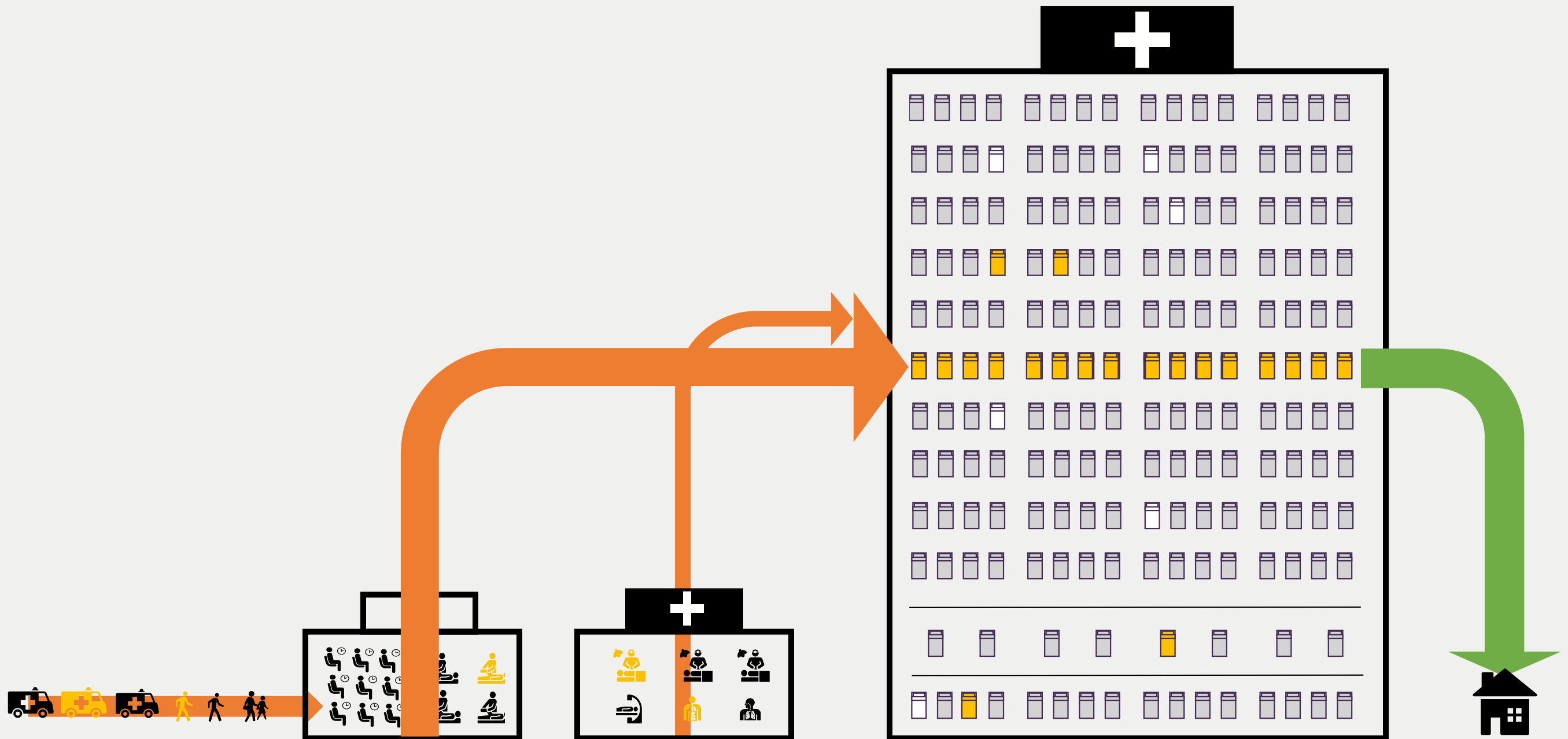
Excludes beds for patients under 18, and patients marked as OTF

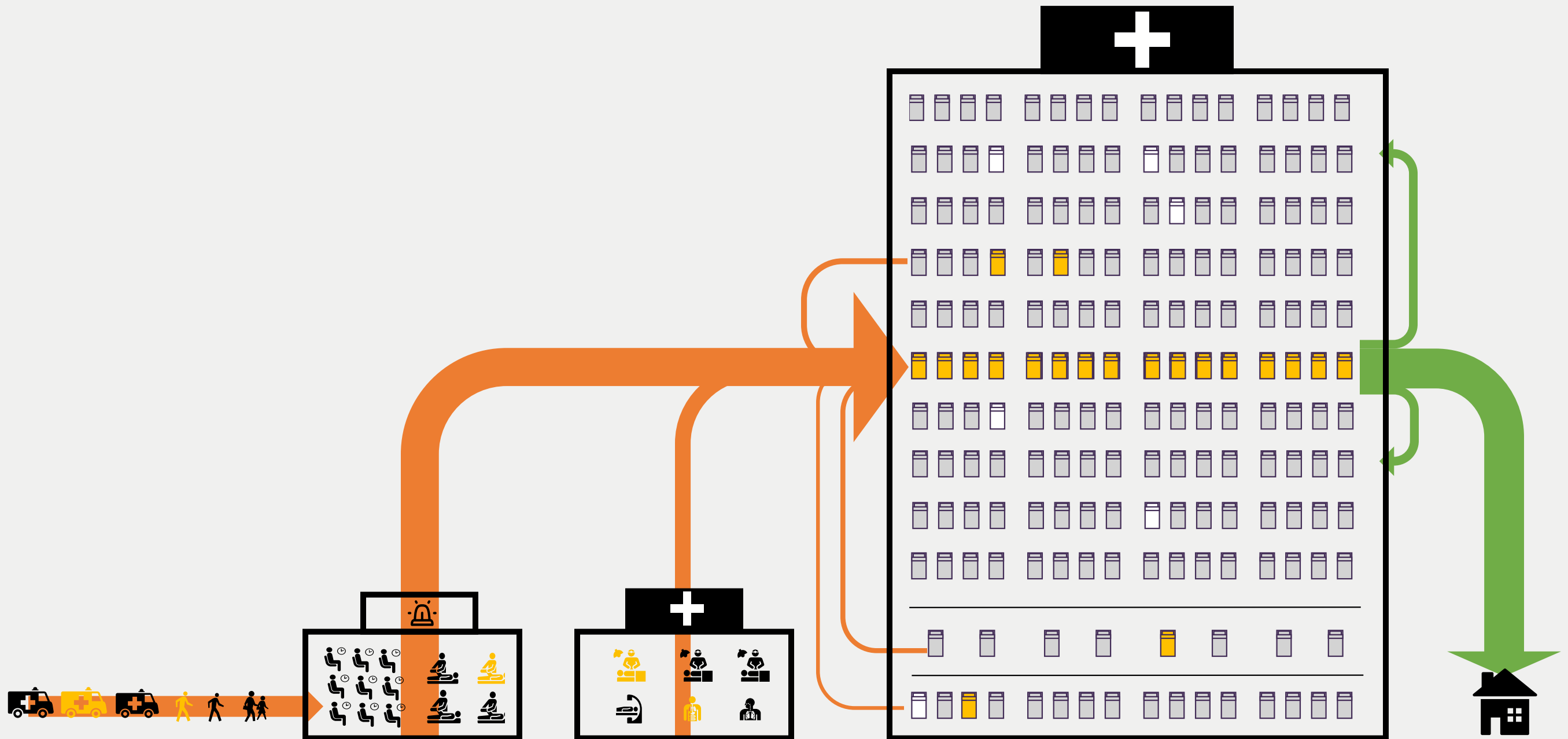


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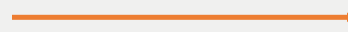




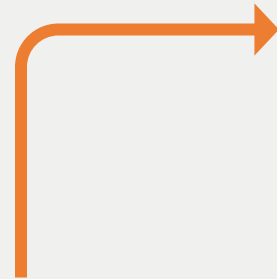
Admissions via ED



Unplanned admissions



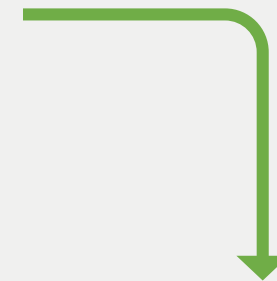
Planned admissions



Incoming from other
specialties

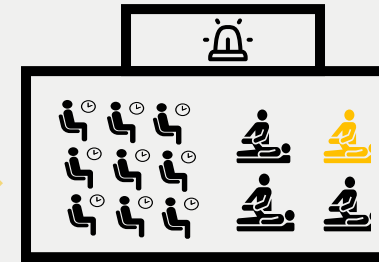
Patients currently in
specialty

Leaving specialty



Outgoing to
another specialty

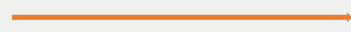
Admissions via ED



XGB Classifier

Mean arrival rate

Unplanned admissions



Mean arrival rate

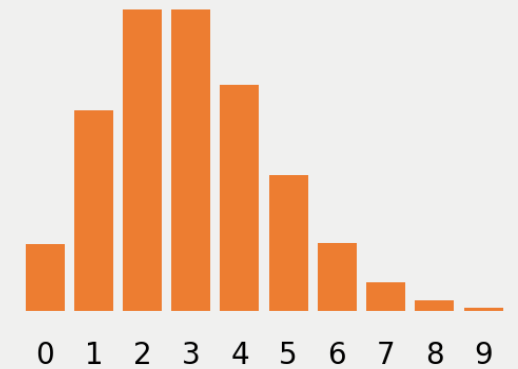
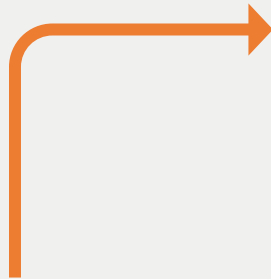
Planned admissions

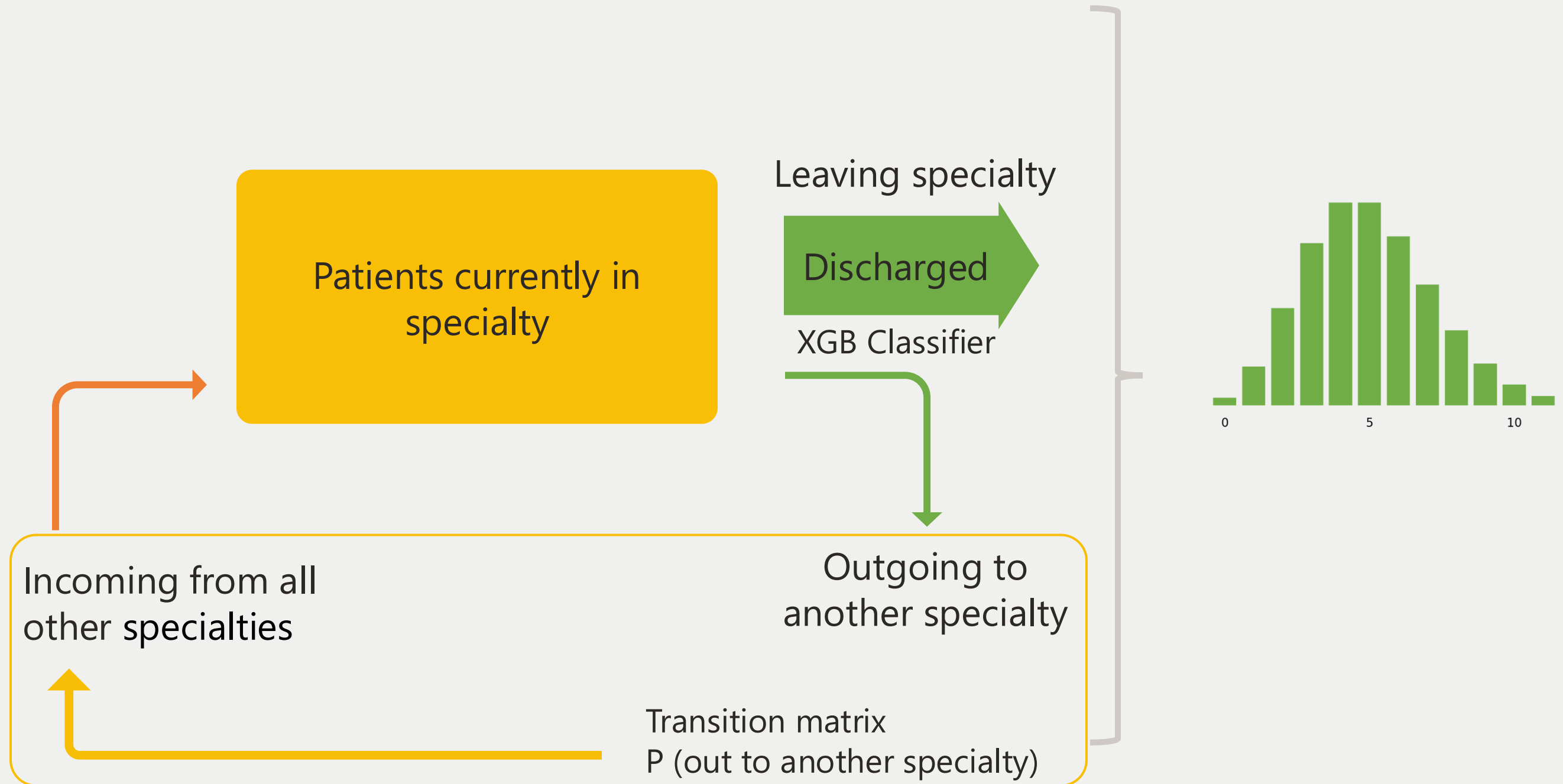


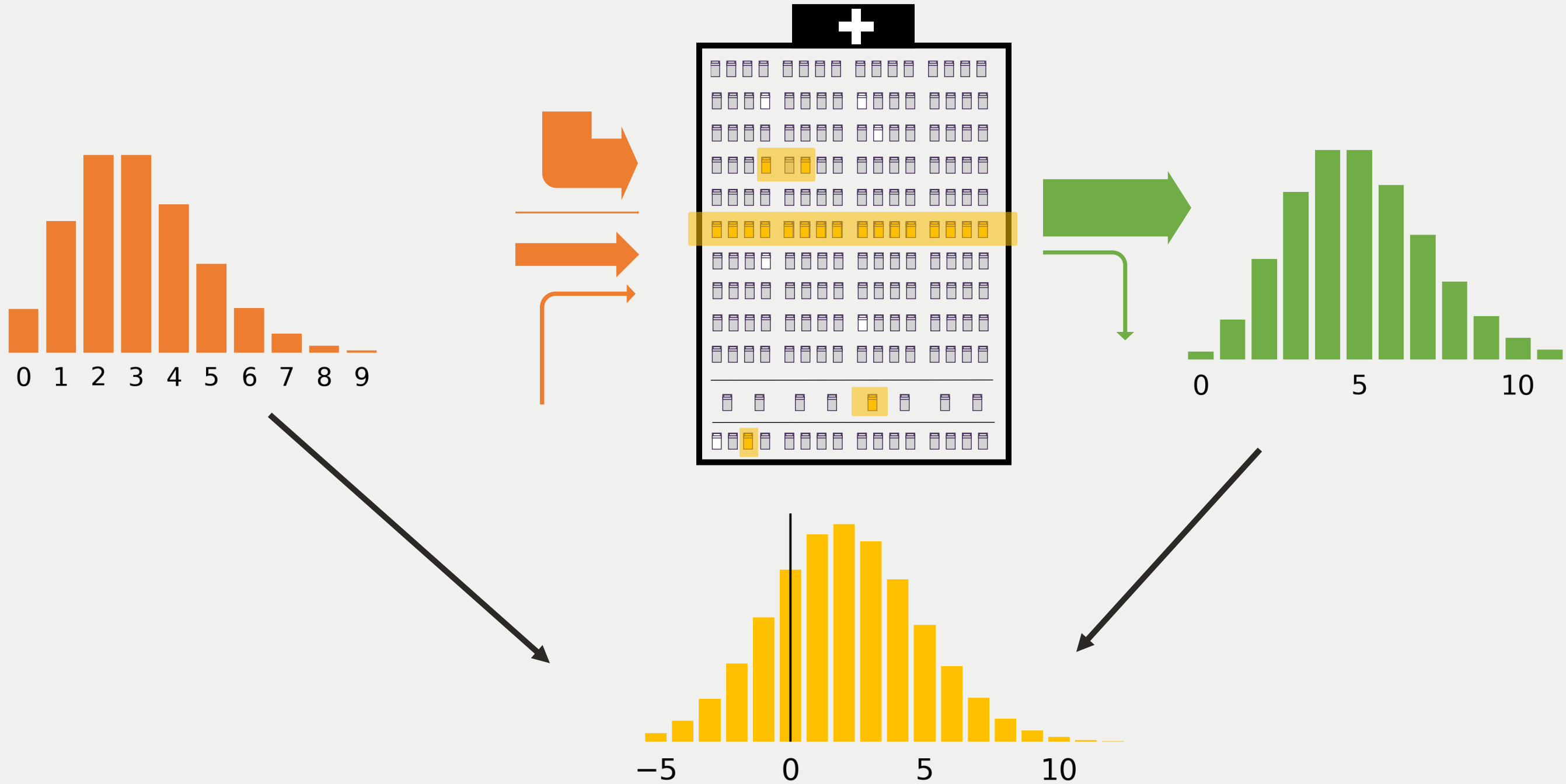
Mean arrival rate

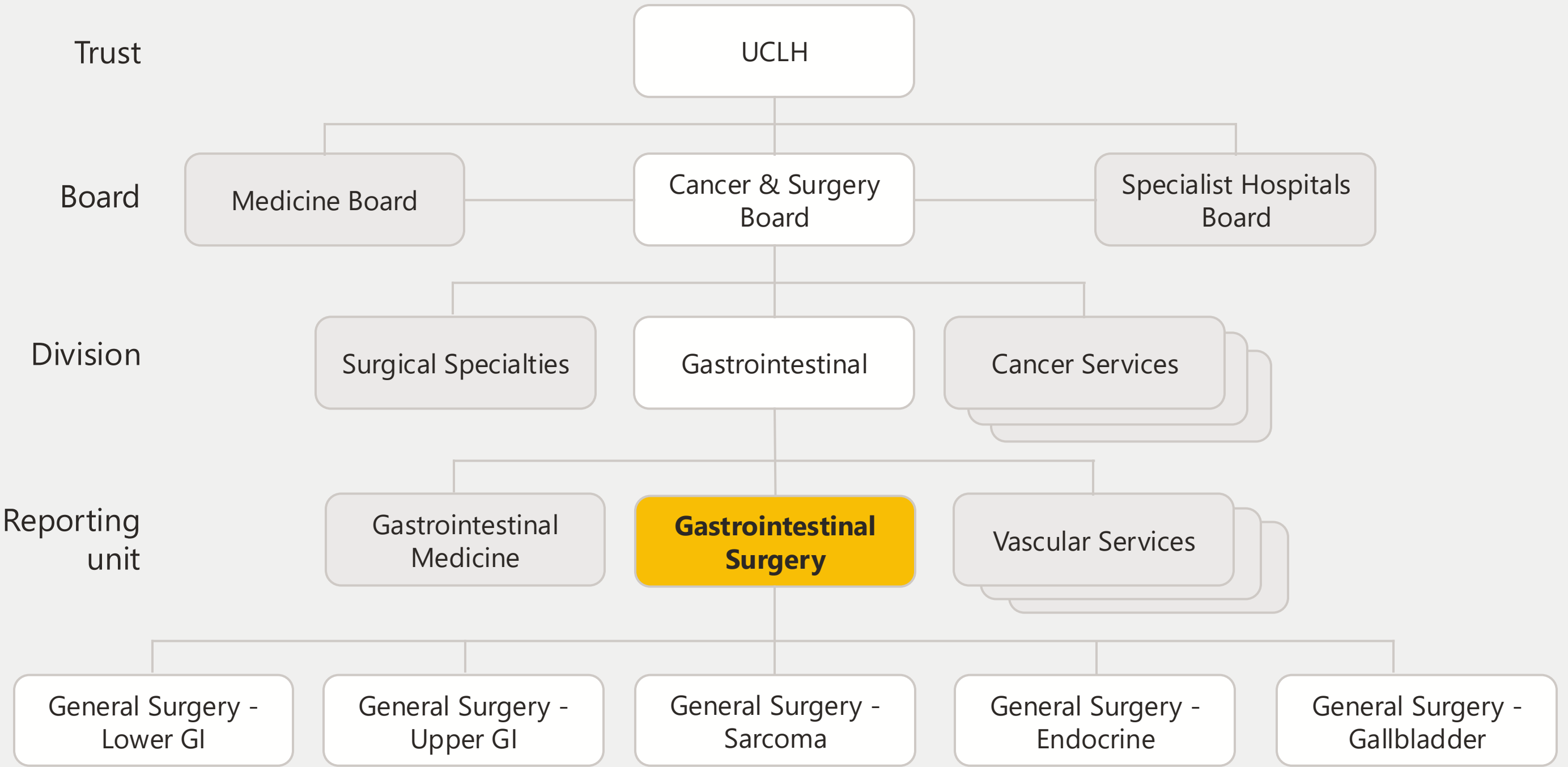
(See next slide)

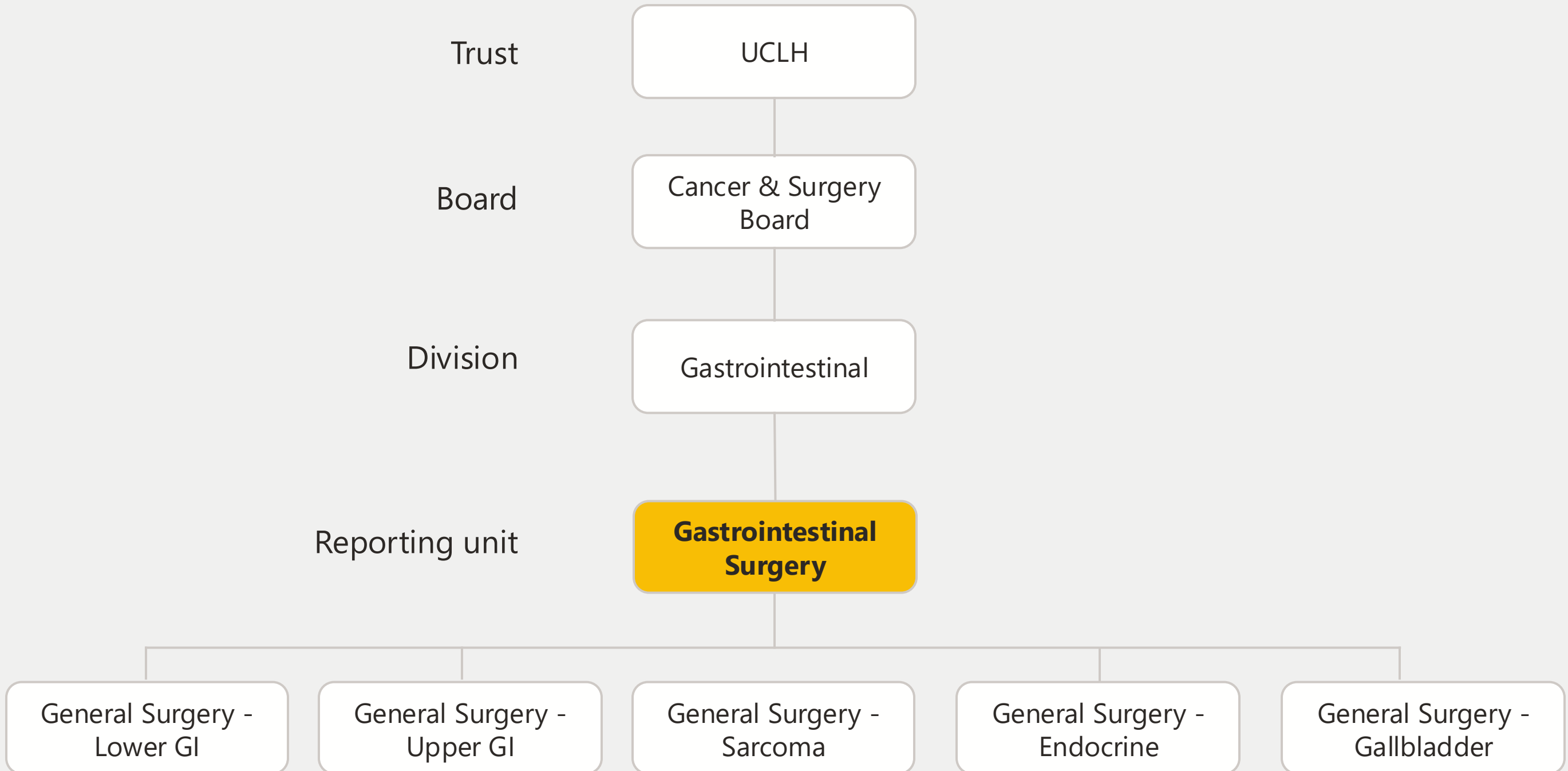
Incoming from other specialties











Reporting unit

Gastrointestinal
Surgery

General Surgery -
Lower GI

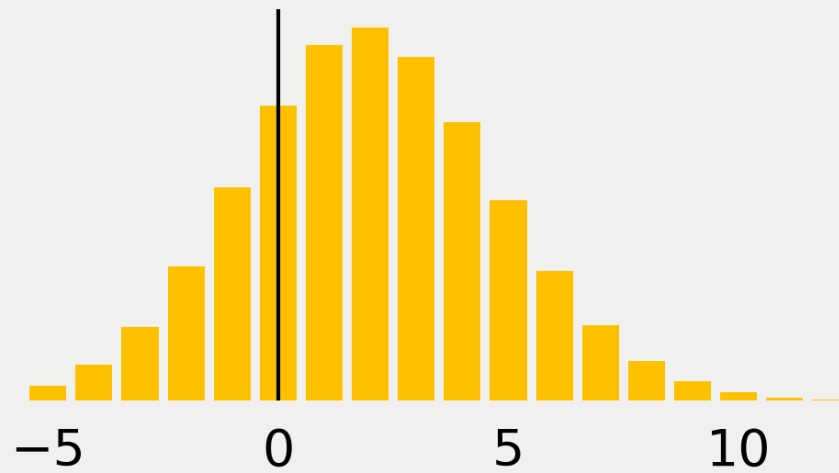
**General Surgery -
Upper GI**

General Surgery -
Sarcoma

General Surgery -
Endocrine

General Surgery -
Gallbladder

Net flow expectation = +1.9



Reporting unit

Gastrointestinal
Surgery

General Surgery -
Lower GI

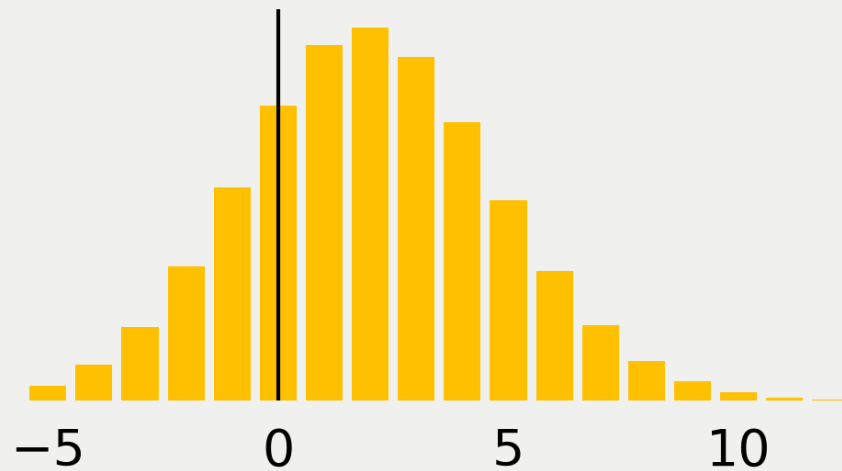
**General Surgery -
Upper GI**

General Surgery -
Sarcoma

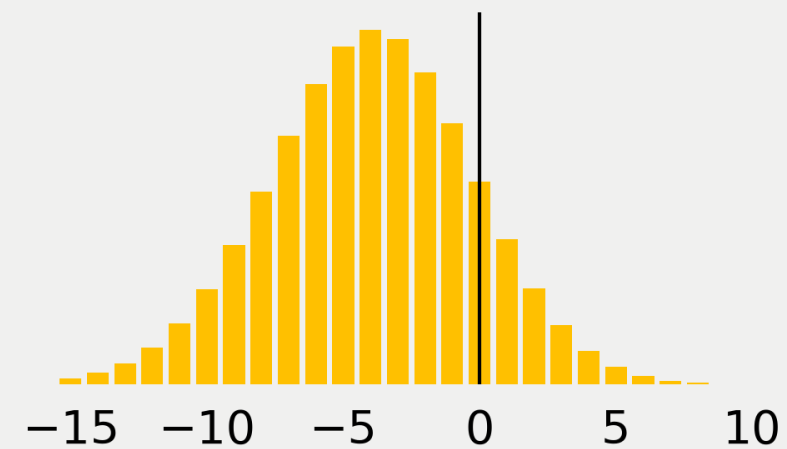
**General Surgery -
Endocrine**

General Surgery -
Gallbladder

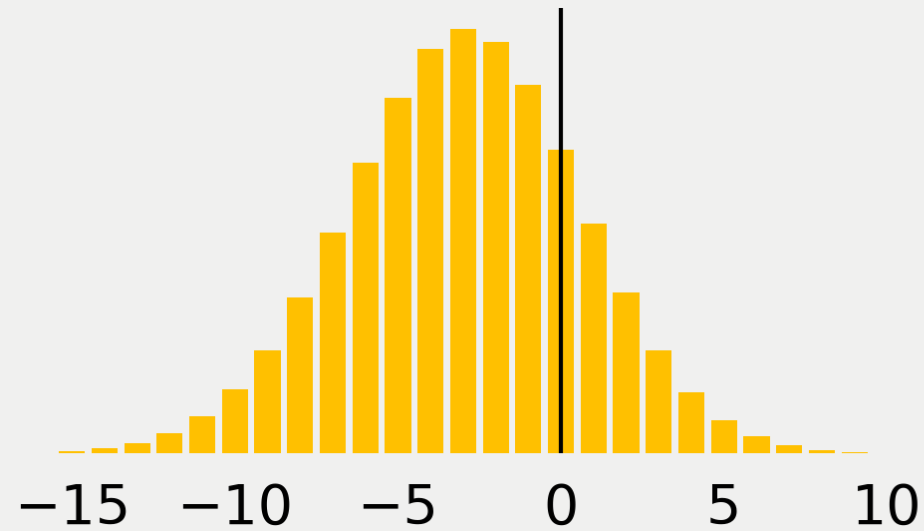
Net flow expectation = +1.9



Net flow expectation = -4.0



Net flow expectation = -3.4



Reporting unit

**Gastrointestinal
Surgery**

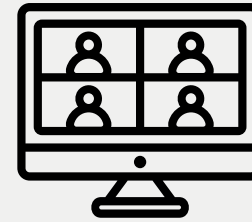
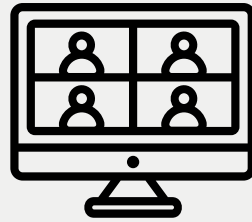
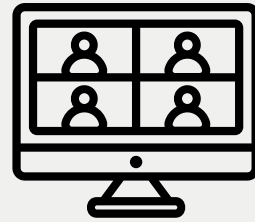
General Surgery -
Lower GI

General Surgery -
Upper GI

General Surgery -
Sarcoma

General Surgery -
Endocrine

General Surgery -
Gallbladder



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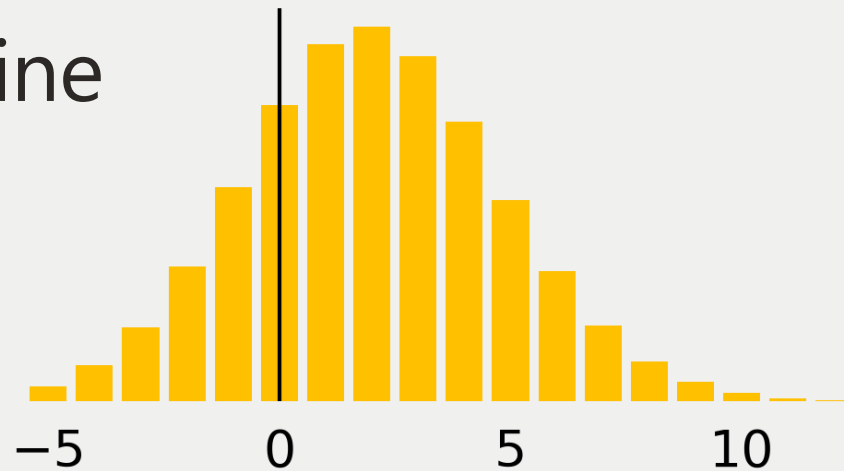
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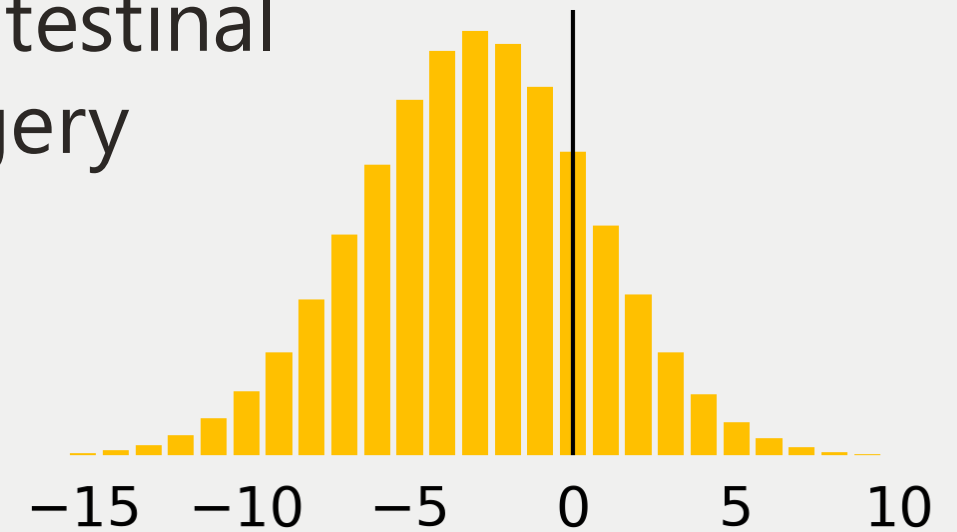
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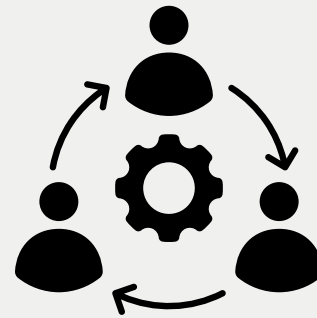
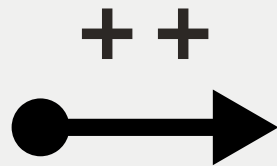
+1

Gastrointestinal
Medicine



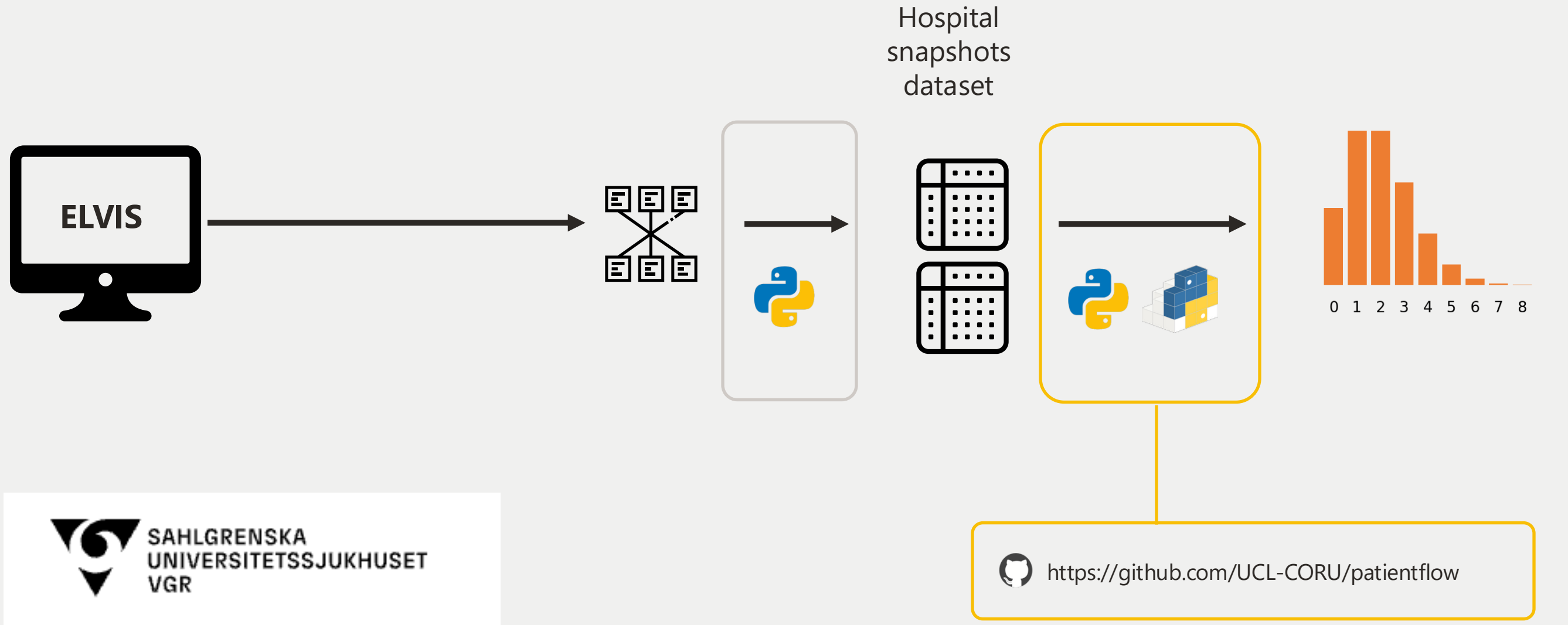
Gastrointestinal
Surgery





Real-time data

Real-time predictions



Real-time prediction. Closing thoughts.

What problems are suited to real-time data?

- What users truly benefit from real-time predictions? What decisions are made as a result?

Modelling with real-time data

- No retrospective coding.
- Unfinished hospital visits. Need “snapshots” of the hospital to train on.

Deploying real-time applications

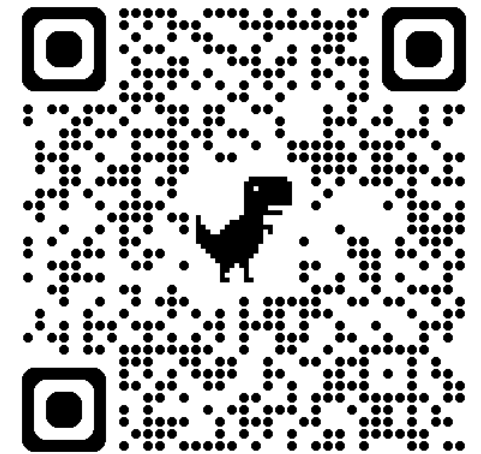
How to evaluate real-time predictions that are intended to change behaviour?

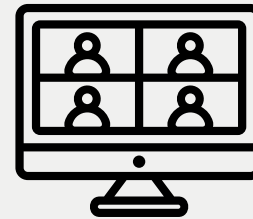
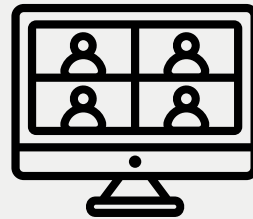
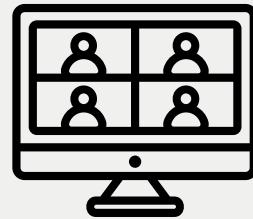
Contact

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Clinical Operational Research Unit
Dept of Mathematics
UCL

zella.king@ucl.ac.uk
zella.king1@nhs.net

patientflow repo





06.00

09.30

12:00

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22:00

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+10

+1

-5

-8

